

Assistive Device to Import Perceptual Ability to Visually-Impaired using Intelligent Scene Captioning



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The major sensory organ of a person is their eyes. One glimpse around us is enough to make us realize the importance of visual information. Most of this information is inaccessible for the blind and visually impaired people which inhibit their independence. Visually impaired people face numerous difficulties in their daily life, and technological interventions may assist them to meet these challenges. Science and engineering can make technical interventions to the lives of visually impaired people in making them independent to navigate and perceive objects around them. Many devices have been proposed to assist a visually challenged person, but most of the devices focus either on object detection through computer vision or on obstacle detection through different sensors, such as GPS, distance sensors, etc. However, the effective utilization of sensor-based technologies and computer vision could result in a highly efficient and supportive device, to make them aware of the surroundings. An artificial intelligence-based fully automatic assistive technology has been developed to recognize different objects, and auditory inputs are provided to the user in real time, which gives better understanding to the visually impaired person about their surroundings.

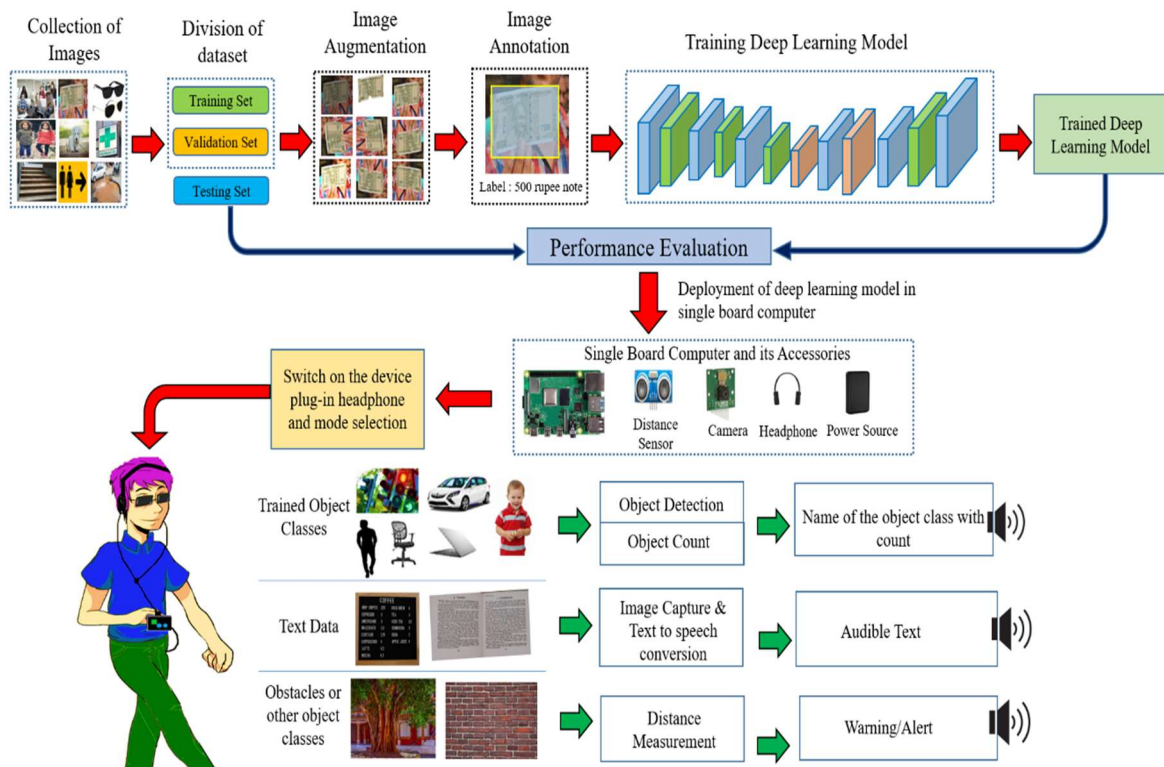


Fig. 1: Block Diagram of Artificial Intelligence based Assistive Framework for Visually Impaired People.

A deep-learning model is trained with various images of objects that are highly relevant to the visually impaired person. Collected images are augmented and manually annotated to bring more robustness to the trained model. In addition to computer vision-based techniques for object recognition, a distance-measuring sensor is integrated to make the device more comprehensive by recognizing obstacles while navigating from one place to another. The auditory information that is conveyed to the user after scene segmentation and obstacle identification is optimized to obtain more information in less time for faster processing of video frames.

The whole framework is standalone and designed for low-cost processors, so that the visually impaired can use it properly without internet connectivity. The detected output is also optimized to ensure faster frame processing and a greater extraction of information i.e., count of objects in a shorter time period. This device can easily differentiate between obstacles and known objects. This methodology can make a significant contribution to assist visually impaired people compared to previously developed methods, which were only focused on obstacle detection and location tracking with the help of basic sensors without use of deep learning. With the use of this methodology, users can interpret more about the things going around and receive obstacle-aware navigation as well.

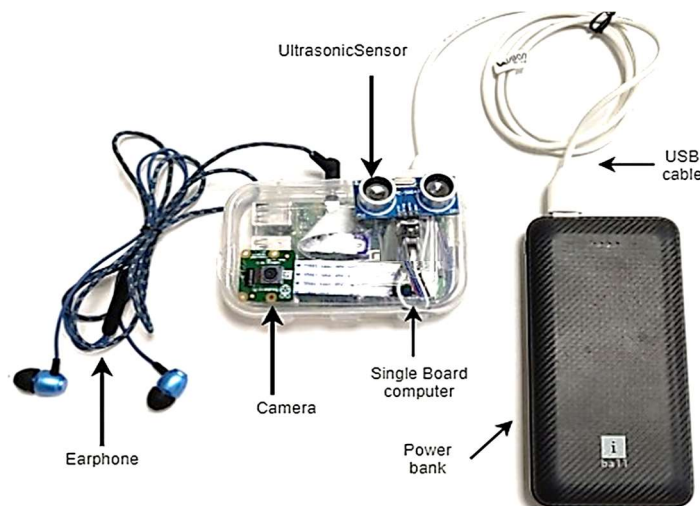


Fig. 2: Developed Prototype

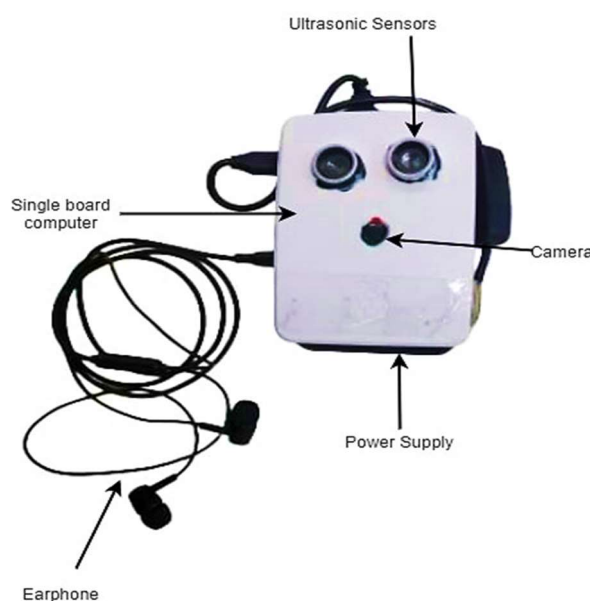


Fig. 3: New modified blind assistive prototype



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This device works on three modes i.e., indoor, outdoor and text reader mode. The indoor mode has less threshold value for the distance of obstacle as compared to outdoor mode. Outdoor mode also has an image enlarge function so that the far objects can be detected early. The text reader mode can be used efficiently where the user has to read such as reading a book, reading a restaurant menu, etc. To read text Optical Character Recognizer (OCR) is used after pre-processing of the input image frame. Optical character recognition or optical character reader (OCR) is the electronic or mechanical conversion of images of typed, handwritten, or printed text into machine-encoded text, whether from a scanned document, a photo of a document, a scene-photo (for example the text on signs and billboards in a landscape photo) or from subtitle text superimposed on an image. It uses artificial intelligence for text search and its recognition of images. For reading text tesseract an optical character recognition engine with open-source code is used which is the most popular and qualitative OCR library. Face recognizer can also be associated with the device where users can identify the known persons and family members which will help them to be social and secure.

The newer version of the prototype is developed in the direction of more efficient and compact device for assistance of visually impaired people which supports multiple languages, works on real time and have more functionalities. This new modified version of a blind assistive device is more handy and easy to handle, easy to carry anywhere.

• Working Process:

- Switch on the device.
- Swipe hand in front of the device to select different modes.
- Device capture image from video streaming.
- Perform object detection and classification using trained model
- Device give audio output of object's name or obstacle.
- Device will read the text on selection of TEXT READER mode.
- On not selecting any mode device will get switched off.

"Disability is only a state of mind. With the use of adaptive equipment and training with it, the mobility can be made easier. Hence, sensors and computer-vision based techniques are integrated to provide convenient a travel-aid to visually impaired people, through which a person can perceive multiple objects, detect obstacles and avoid collisions." - Prof. M.K. Dutta (Principal Investigator)



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- **Patents from the project:**

“Multi Mode Real-Time Object Detector and Obstacle Aware Assistive System for Visually Impaired Person (VIP)”, Application No.- 201911047766, Dated - 22.11.2019, Status – Applied and Under Examination.

- **Publications from the project:**

1. Joshi, R.C.; Yadav, S.; Dutta, M.K.; Travieso-Gonzalez, C.M. Efficient Multi-Object Detection and Smart Navigation Using Artificial Intelligence for Visually Impaired People. Entropy 2020, 22, 941.
2. Yadav, R. C. Joshi, M. K. Dutta, Martin Kiac, Pavel Sikora, “Fusion of Object Recognition and Obstacle Detection approach for Assisting Visually Challenged Person” 2020 43rd International Conference on Telecommunications and Signal Processing (TSP), July 7-9, 2020, Milan, Italy.
3. R. C. Joshi, S. Yadav and M. K. Dutta, "YOLO-v3 Based Currency Detection and Recognition System for Visually Impaired Persons," 2020 International Conference on Contemporary Computing and Applications (IC3A), Lucknow, India, 2020.
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5. Lalita Jaiswar, Anjali Yadav, Malay Kishore Dutta, Carlos Travieso-González, Luis Esteban-Hernández, “Transfer Learning based Computer Vision Technology for Assisting Visually Impaired for detection of Common Places” 3rd International Conference on Applications of Intelligent Systems 2020 (APPIS 2020) Las Palmas de Gran Canaria, Spain, ACM Digital Library. DOI: 10.1145/3378184.3378215
6. Saumya Yadav, R. C. Joshi, M. K. Dutta, R. Burget. “Artificial Intelligence based Robust and Real-time Assistance for Visually Impaired People”- Assistive Technology (The Official Journal of RESNA), Taylor & Francis Publishers- Submitted and under review.